Mark schemes

Q1.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct vector	AO1.1b	B1	$\begin{pmatrix} -4 \\ -3 \\ 6 \end{pmatrix}$
(b)	Obtains one other edge as vector	AO1.1a	M1	$\overrightarrow{BC} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}$
	Obtains $\overrightarrow{DC}$ correctly Or Obtains correctly both $\overrightarrow{BC}$ and $\overrightarrow{AD}$ Or Obtains correctly both $\overrightarrow{CB}$ and $\overrightarrow{DA}$	AO1.1b	A1	$\overrightarrow{AD} = \begin{pmatrix} 1 \\ 5 \\ -1 \end{pmatrix}$ $\overrightarrow{DC} = \begin{pmatrix} -4 \\ -3 \\ 6 \end{pmatrix}$
	Obtains length of one edge (or its square)	AO1.1a	M1	$AB = \sqrt{(-4)^2 + (-3)^2 + 6^2}$ $= \sqrt{61}$
	Obtains two correct lengths of different edges	AO1.1b	A1	$AD = \sqrt{1^2 + 5^2 + (-1)^2}$ $= 3\sqrt{3}$
	Completes rigorous argument to show *ABCD* is a parallelogram and not a rhombus	AO2.1	R1	$\overrightarrow{AB} = \overrightarrow{DC}$ *ABCD* must be a parallelogram $AB \neq AD$ *ABCD* is not a rhombus
				Total 6 marks

Q2.

	Marking Instructions	AO	Marks	Typical Solution
(a)(i)	Sums the forces given correctly	AO1.1b	B1	$\mathbf{F} = \mathbf{F}_1 + \mathbf{F}_2 + \mathbf{F}_3$ $= 33\mathbf{i} - 11\mathbf{j}$
	Uses Pythagoras to find the magnitude of the vector and obtains correct magnitude (given to 3 sig	AO1.1b	B1	$\lvert \mathbf{F} \rvert = \sqrt{33^2 + (-11)^2}$ $= 34.8\ \text{N (3 sf)}$

	figs)			
(ii)	Uses trig expression with appropriate values	AO1.1a	M1	$\tan\theta = \frac{11}{33}$
	Obtains correct angle (given to nearest 0.1°)	AO1.1b	A1	$\theta = \tan^{-1}\left(\frac{1}{3}\right)$ = 18.4° (3 sf) **OR** $\sin\theta = \frac{11}{\sqrt{1210}}$ $\theta = \sin^{-1}\left(\frac{11}{\sqrt{1210}}\right)$ = 18.4° (3 sf) **OR** $\cos\theta = \frac{33}{\sqrt{1210}}$ $\theta = \cos^{-1}\left(\frac{33}{\sqrt{1210}}\right)$ = 18.4° (3 sf)
(b)	States negative of 'their' part **(a)(i)**	AO2.2a	B1F	$\mathbf{F}_4 = -33\mathbf{i} + 11\mathbf{j}$
				Total 5 marks

Q3.

	Marking Instructions	**AO**	**Marks**	**Typical Solution**
(a)	Integrates both components with at least one correct	AO1.1a	M1	$\mathbf{r} = \int 40e^{-0.2t}\,dt\,\mathbf{i} + \int 50(e^{-0.2t} - 1)\,dt\,\mathbf{j}$ $= (-200e^{-0.2t} + c)\mathbf{i} + (-250e^{-0.2t} - 50t + d)\mathbf{j}$
	Obtains correct terms. (condone missing constants)	AO1.1b	A1	$t = 0$, $\mathbf{r} = 0\mathbf{i} + 0\mathbf{j} \Rightarrow c = 200$, $d = 250$ **OR**
	Evaluates both constants (or uses definite integration) using 'their' expression for **r**	AO3.4	M1	$\mathbf{r} = \int_0^t 40e^{-0.2t}\,dt\,\mathbf{i} + \int_0^t 50(e^{-0.2t} - 1)\,dt\,\mathbf{j}$ $= \left[-200e^{-0.2t}\mathbf{i} + (-250e^{-0.2t} - 50t)\mathbf{j}\right]_0^t$
	Obtains correct expression	AO1.1b	A1	$\mathbf{r} = 200(1 - e^{-0.2t})\mathbf{i}$ $+ (250 - 250e^{-0.2t} - 50t)\mathbf{j}$

(b)	Forms equation to find t based on horizontal component	AO3.4	M1	$200(1-e^{-0.2t}) = 100$ $t = 5\ln 2 = 3.4657$ $y = 250 - 250e^{-0.2\times 5\ln 2} - 50\times 5\ln 2$ $= -48.3$
	Obtains correct time	AO1.1b	A1	
	Substitutes 'their' time into vertical component	AO1.1a	M1	
	Obtains correct displacement for 'their' time (to 1 sf, must have metres) FT only if both M1 marks have been awarded	AO3.2a	A1F	The parachutist has a vertical displacement of 50 m below the origin
(c)	Identifies vertical component of the velocity as first step	AO2.4	M1	$\frac{\mathrm{d}}{\mathrm{d}t}(50(e^{-0.2t} - 1)) = -10e^{-0.2t}$ As there is no initial vertical component of velocity (and hence no air resistance) the initial acceleration is only due to gravity. Hence g is taken as 10 m s^{-2}
	Differentiates vertical component of velocity correctly	AO1.1b	A1	
	Considers implication of initial motion and reaches correct conclusion	AO2.2a	R1	
				Total 11 marks

Q4.

	Marking Instructions	AO	Marks	Typical Solution
(a)	Obtains correct horizontal component of the initial velocity	AO1.1b	B1	$2.5U = 40$ $U = 16$
	Forms equation to find vertical component of initial velocity	AO3.3	M1	$-10 = 2.5V - 0.5 \times 9.81 \times 2.5^2$
	Obtains correct vertical component of initial velocity	AO1.1b	A1	$V = 8.2625$
	Forms equation for vertical component of velocity at height 3 using 'their' derived values for U and V	AO3.4	M1	$v_y^2 = 8.2625^2 + 2\times(-9.81)\times 3$
	Obtains correct component of velocity	AO1.1b	A1	$v_y = 3.067\ldots$

	Correct final speed with units, correct for 'their' U and v_y FT applies only if both M1 marks have been awarded	AO3.2a	A1F	$v = \sqrt{16^2 + 3.067^2} = 16.3\text{ m s}^{-1}$
(b)	States 'their' value of horizontal component of the initial velocity from part **(a)**	AO3.4	A1F	16 m s $^{-1}$
(c)	Explains that horizontal velocity has been assumed to be constant in their model and that this is not likely to be true, with valid reasoning	AO3.5b	E1	It was assumed that there were no resistance forces acting on the ball which is unlikely to be true in reality. The horizontal speed of the ball is likely to vary… air resistance would slow the ball down, wind might speed the ball up
				Total 8 marks

Q5.

(a) $v = \int a \, dt$

$= (20t^2 + t^3)\mathbf{i} - 5e^{-4t}\,\mathbf{j} + \mathbf{c}$

M1 for either term correct.
*Condone no '+ **c**'.*

M1A1

When $t = 1$,
$6\mathbf{i} - 5e^{-4}\mathbf{j} = 21\mathbf{i} - 5e^{-4}\mathbf{j} + \mathbf{c}$

*Finding '+ **c**'; not using **c** = 6**i** – 5e^{-4} **j**.*

M1

$\mathbf{c} = -15\mathbf{i}$

A1

$\mathbf{v} = (20t^2 + t^3 - 15)\mathbf{i}\ 5e^{-4t}\,\mathbf{j}$

A1

5

(b) When $t = 0$, $\mathbf{v} = -15\mathbf{i} - 5\mathbf{j}$

M1

Speed is $\sqrt{15^2 + 5^2}$

M1

= 15.8 m s^{-1}

Accept 5 $\sqrt{10}$

A1

3

[8]

Q6.

(a) using $\mathbf{F} = m\mathbf{a}$:

ie dividing by 50

M1

$\mathbf{a} = (6t - 1.2\,t^2)\,\mathbf{i} + 2\,e^{-2t}\,\mathbf{j}$

A1

2

(b) $v = \int \mathbf{a}\,dt$

$= (3\,t^2 - 0.4\,t^3)\,\mathbf{i} - e^{-2t}\mathbf{j} + \mathbf{c}$

when $t = 0$, $\mathbf{r} = 7\,\mathbf{i} - 4\,\mathbf{j}$

condone lack of + c; M1 one term correct

M1A1

$\mathbf{c} = 7\,\mathbf{i} - 3\mathbf{j}$

$\mathbf{v} = (7 + 3\,t^2 - 0.4\,t^3)\,\mathbf{i} - (3 + e^{-2t})\,\mathbf{j}$

ft from ke^{-2t} *in (b); just adding 7****i*** *– 4****j****, m0 accept unsimplified. CAO*

m1A1

4

(c) when $t = 1$, $\mathbf{v} = 9.6\,\mathbf{i} - 3.135\,\mathbf{j}$

ft from (b)

M1A1

speed = $\sqrt{9.6^2 + 3.135^2}$

m1

= 10.1 ms $^{-1}$

ft from (b)

A1

4

[10]

Q7.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\boldsymbol{r}$*.*

A1: Correct expression for $\boldsymbol{r}$*. Allow equivalent column vector answer.*

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\boldsymbol{j}$ *component equal to zero to form a quadratic equation.*

A1: Correct equation.

M1A1

$t = 30$ seconds

A1: For 30 seconds. No need to see $t = 0$*.*

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t*. Can be implied by subsequent working in terms of* t*.*

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$*. May be with their components if velocity stated incorrectly.*

A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases
If the equation in t in line 2 is not seen:
then seeing t = 20 and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ and v = 1.41 award 4 out of 6
or
then seeing t = 20 and $\boldsymbol{v} = \boldsymbol{i} - \boldsymbol{j}$ award 2 out of 6

6

[11]

Q8.

(a) $\mathbf{r} = (-\mathbf{i} + 3\mathbf{j})t + \frac{1}{2}(0.1\mathbf{i} - 0.2\mathbf{j})t^2$

M1: Using constant acceleration equation to get $\boldsymbol{r}$.

A1: Correct expression for $\boldsymbol{r}$. Allow equivalent column vector answer.

M1A1

2

(b) $3t - 0.1t^2 = 0$

$t(3 - 0.1t) = 0$

$t = 0$ or $t = 30$

M1: Putting their $\boldsymbol{j}$ component equal to zero to form a quadratic equation.
A1: Correct equation.

M1A1

t = 30 seconds

A1: For 30 seconds. No need to see t = 0.

A1

3

(c) $\mathbf{v} = (0.1t - 1)\mathbf{i} + (3 - 0.2t)\mathbf{j}$

B1: Correct expression for the velocity in terms of t.
Can be implied by subsequent working in terms of t.

B1

$0.1t - 1 = -(3 - 0.2t)$

$2 = 0.1t$

M1: For $0.1t - 1 = \pm(3 - 0.2t)$. May be with their components if velocity stated incorrectly.
A1: Correct equation.

M1A1

$t = 20$

A1: $t = 20$

A1

$\mathbf{v} = \mathbf{i} - \mathbf{j}$

$v = \sqrt{2} = 1.41 \text{ ms}^{-1}$

dM1: finding velocity and speed at their time

dM1

A1: Correct speed.

A1

Special cases
If the equation in t in line 2 is not seen:
then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ and $v = 1.41$ award 4 out of 6
or
then seeing $t = 20$ and $\mathbf{v} = \mathbf{i} - \mathbf{j}$ award 2 out of 6

6

[11]

Q9.

(a) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j})t$

*M1: Use of constant acceleration equation to find **v** with $\mathbf{u} \neq 0\mathbf{i} + 0\mathbf{j}$*
*A1: Correct **v**.*
(Could be done as a column vector.)

M1A1

2

(b) (i) $\mathbf{v} = (4\mathbf{i} + 0.5\mathbf{j}) + (-0.4\mathbf{i} + 0.2\mathbf{j}) \times 22.5$

M1: Substitution of 22.5 into their expression for the elocity, even if no marks awarded in part (a).

M1

$= -5\mathbf{i} + 5\mathbf{j}$

A1: Correct velocity CAO
(Could be done using column vectors.)

A1

2

(ii) North-west

B1: Correct statement of direction. Accept 315°.
Must follow from correct answer to (b)(i).

B1

1

(c) $(\mathbf{v} =)(4 - 0.4t)\mathbf{i} + (0.5 + 0.2t)\mathbf{j}$

*B1: Grouping **i** and **j** components at some point in the solution.*
(Could be done using column vectors.)
*Allow 5 = (4 – 0.4t)**i** + (0.5 + 0.2t)**j***

B1

$5^2 = (4 - 0.4t)^2 + (0.5 + 0.2t)^2$

M1: Seeing both components of their velocity squared and added

*A1: Correct equation. (Condone including **i** and **j**.)*
For example: $5 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A0
$5^2 = (4 - 0.4t)\mathbf{i}^2 + (0.5 + 0.2t)\mathbf{j}^2$ scores B1M1A1

M1A1

$0.2t^2 - 3t - 8.75 = 0$

A1: Any correct simplified quadratic equation, with exactly three terms.

A1

$t^2 - 15t - 43.75 = 0$

$t = 17.5$ or $t = -2.5$

dM1: Solving the quadratic equation.

(Allow one substitution error in correctly quoted formula)
Candidates with an incorrect quadratic equation must show method to get dM1.

dM1

$t = 17.5$

A1: Correct positive solution stated.

A1

6

[11]

Q10.

(a) $\mathbf{r} = \int v\,\mathrm{d}t = (4t + t^3)\mathbf{i} + (12t - 4t^2)\mathbf{j} + \mathbf{c}$

M1: either ***i*** *or* ***j*** *term correct.*
Condone no ***c***

M1A1

When $t = 0$, $\mathbf{r} = 5\mathbf{i} - 7\mathbf{j}$
$\mathbf{c} = 5\mathbf{i} - 7\mathbf{j}$

Any attempt at ***c***

M1

$\mathbf{r} = (5 + 4t + t^3)\mathbf{i} + (-7 + 12t - 4t^2)\mathbf{j}$

A1

4

(b) $\mathbf{a} = \dfrac{\mathrm{d}v}{\mathrm{d}t}$

$\mathbf{a} = 6t\,\mathbf{i} - 8\,\mathbf{j}$

M1: either term correct

M1A1

2

(c) Using $\mathbf{F} = m\mathbf{a}$

Or: using $\boldsymbol{F} = m\boldsymbol{a}$

M1

$\mathbf{F} = 2(6t\,\mathbf{i} - 8\,\mathbf{j})$

$\boldsymbol{F} = 2(6t\,\boldsymbol{i} - 8\,\boldsymbol{j})$

$= 12t\,\mathbf{i} - 16\,\mathbf{j}$

When $t = 1$, $\boldsymbol{F} = 12\,\boldsymbol{i} - 16\,\boldsymbol{j}$

A1

$\therefore$ Magnitude of force is $(144t^2 + 256)^{\frac{1}{2}}$ when $t = 1$

Magnitude of force is $(144 + 256)^{\frac{1}{2}}$

M1

$= 20$ N

= 20 N

A1

4

[10]

Q11.

(a) **v** = (0.5**i** + 0.375**j**) × 20 (=10**i** + 7.5**j**)

*M1: Calculating velocity with **u** = 0**i** + 0**j** and t = 20.*

A1: Correct expression for velocity.

M1A1

$$v = \sqrt{10^2 + 7.5^2} = 12.5 \text{ ms}^{-1}$$

dM1: Calculating speed.

A1: Correct speed.

dM1A1

4

(b) $$\tan\theta = \frac{0.5}{0.375} \text{ or } \frac{10}{7.5} \left(\text{or } \tan\theta = \frac{0.375}{0.5} \text{ or } \frac{7.5}{10} \right)$$

M1: Using trig to find angle. Can be inverted as shown in brackets.

A1F: Correct trig expression with any correct equivalent fraction.

M1A1F

θ = 053°

A1: Correct angle to the nearest degree. Accept 53°.

A1

OR

$$\cos\theta = \frac{7.5}{12.5} \text{ or } \frac{0.375}{0.625} \left(\text{or } \cos\theta = \frac{10}{12.5} \right)$$

(M1A1F)

θ = 053°

Note: For 37° award M1A0A0
But for 90 – 37 = 53° award M1A1A1.
For 127°, award M1A1A0

(A1)

OR

$$\sin\theta = \frac{10}{12.5} \text{ or } \frac{0.5}{0.625} \left(\text{or } \sin\theta = \frac{7.5}{12.5} \right)$$

Note: 53.1° as final answer scores M1A1A0

(M1A1F)

θ = 053°

Condone finding angle from acceleration or position vector.

(A1)

3

(c) (**r** =) $\frac{1}{2}$ (0.5**i** + 0.375**j**) t^2 (= 0.25 t^2 **i** + 0.1875t^2 **j**)

M1: Finding an expression for position vector in terms of t.

A1: Correct position vector.

M1A1

$500^2 = (0.25\,t^2)^2 + (0.1875\,t^2)^2$

dM1: Using distance to form an equation for t.

A1: Correct equation.

dM1A1

$$t = 4\sqrt{\frac{500^2}{0.25^2 + 0.1875^2}} = 40 \text{ seconds}$$

A1: Correct time.

A1

OR

a = 0.625

M1: Finding magnitude of acceleration.

A1: Correct acceleration

(M1A1)

$$500 = \frac{1}{2}0.625t^2$$

dM1: Using distance to form an equation for t.

A1: Correct equation.

(dM1A1)

t = 40

A1: Correct time.

(A1)

OR

$400 = \frac{1}{2}\,0.5t^2$ or $300 = \frac{1}{2}\,0.375t^2$

M1: Working with one component.

A1: Correct distance (300 or 400)

A1: Correct equation.

(M1A1)
(A1)

$t^2 = 1600$

dM1: Solving for t.

(dM1)

$t = 40$

A1: Correct t.

Note: 500 ÷ 12.5 = 40 is not acceptable and scores 0

(A1)

5

[12]

Q12.

(a) $\mathbf{a} = \frac{\mathrm{d}v}{\mathrm{d}t}$

$\mathbf{a} = -8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}$

*M1: Differentiating with either of the two components correct. Do not need to see **i** or **j**.*

*A1: Correct **i** component.*

*A1: Correct **j** component.*

M1
A1
A1

3

(b) (i) Using $\mathbf{F} = m\mathbf{a}$

$\mathbf{F} = 5 \times \{-8e^{-2t}\,\mathbf{i} + (6 - 6t)\mathbf{j}\}$

$= -40e^{-2t}\,\mathbf{i} + (30 - 30t)\mathbf{j}$

M1: Multiplying their acceleration by 5, even if not a vector.

A1: Correct expression.

M1
A1

2

(ii) Magnitude of $\mathbf{F}$ is

$\{(-40)^2 + (30)^2\}^{\frac{1}{2}}$

M1: Finding magnitude from two non-zero terms. Must *add terms and square root. Condone* $\{(40)^2 + (30)^2\}^{\frac{1}{2}}$

M1

$= 50$

A1: Correct answer only.

In this part, condone lack of negative signs in expression for force in (b) (i).

A1

2

(c) When $\mathbf{F}$ acts due west, $\mathbf{j}$ component is zero

$30 - 30t = 0$

M1: Putting $\mathbf{j}$ *component equal to zero.*

M1

$t = 1$

A1: Correct time.

A1

2

(d) $\mathbf{r} = -2e^{-2t}\,\mathbf{i} + (3t^2 - t^3)\mathbf{j} + \mathbf{c}$

M1: Integration with either of the two components correct. Do not need to see $\mathbf{i}$ *or* $\mathbf{j}$*.*

A1: Correct $\mathbf{i}$ *component.*

A1: Correct $\mathbf{j}$ *component.*

Condone lack of + $\mathbf{c}$

M1
A1
A1

When $t = 0$, $\mathbf{r} = 6\mathbf{i} + 5\mathbf{j}$ $\therefore \mathbf{c} = 8\mathbf{i} + 5\mathbf{j}$

dM1: Finding $\mathbf{c}$ *using* $6\mathbf{i} + 5\mathbf{j}$ *and* $e^0 = 1$*.*

dM1

$\therefore \mathbf{r} = (8 - 2e^{-2t})\mathbf{i} + (5 + \frac{1}{4}t^4 - t^3)\mathbf{j}$

A1: Correct position vector.

A1

5

[14]

Q13.

(a) $\mathbf{v} = 4.5\mathbf{i} + 2\mathbf{j}$

B1: Correct velocity

B1

1

(b) $\mathbf{v} = 4.5\mathbf{j}$

B1: Correct velocity

B1

1

(c) Due North

B1: Correct direction

B1

1

(d) 4.5**j** = 4.5**i** + 2**j** + 5**a**

*M1: Use of a constant acceleration equation to find **a***
A1: Correct equation

M1A1

$$\mathbf{a} = \frac{-4.5\mathbf{i} + 2.5\mathbf{j}}{5} = -0.9\mathbf{i} + 0.5\mathbf{j}$$

A1: Correct acceleration

A1

3

(e) **F** = 250 (–0.9**i** + 0.5**j**) = – 225**i** + 125**j**

M1: Using Newton's second law.

M1

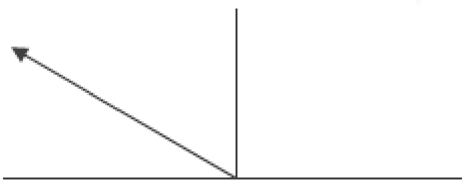
$$F = \sqrt{225^2 + 125^2} = 257\text{ N}$$

M1: Finding magnitude.
A1: Correct magnitude of the force

M1A1

3

(f)



B1: Diagram with arrow in the correct quadrant.

B1

$$\tan^{-1}\left(\frac{125}{225}\right) = 29.1°$$

M1: Use of trig to find an angle.

M1

A1: Correct angle from working.

A1

OR

$\tan^{-1}\left(\frac{0.5}{0.9}\right) = 29.1°$

$180 - 29.1 = 151°$

A1: Finding required angle correctly.

A1

4

[13]

Q14.

(a) $(8\mathbf{i} + 12\mathbf{j}) + (4\mathbf{i} - 4\mathbf{j}) = 12\mathbf{i} + 8\mathbf{j}$

M1: Adding forces to find resultant.
A1: Correct resultant force.

M1A1

2

(b) $4\mathbf{a} = 12\mathbf{i} + 8\mathbf{j}$ or ($\mathbf{a}$ =) $\frac{12\mathbf{i} + 8\mathbf{j}}{4}$

*M1: Use of Newton's second law with 4**a** and their answer to part (a).*

M1

($\mathbf{a}$ =) $3\mathbf{i} + 2\mathbf{j}$ **AG**

A1: Correct acceleration from correct equation.

A1

2

(c) (i) $40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + (3\mathbf{i} + 2\mathbf{j}) \times 20$

$40\mathbf{i} + 32\mathbf{j} = \mathbf{v} + 60\mathbf{i} + 40\mathbf{j}$

B1: Seeing 60i + 40j or (3i + 2j) × 20

B1

*M1: Use of constant acceleration equation with t = 20 and **a** = 3**i** ± 2**j**.*

M1

$\mathbf{v} = (40\mathbf{i} + 32\mathbf{j}) - (60\mathbf{i} + 40\mathbf{j})$

AG

$= -20\mathbf{i} - 8\mathbf{j}$

A1: Correct velocity from correct working, with one of the intermediate lines of working (or equivalent) shown.

Note: Candidates may use ***u*** *instead of* ***v*** *in their working.*
Example:
Starting with
$\boldsymbol{v} = 40\boldsymbol{i} + 32\boldsymbol{j} + (3\boldsymbol{i} + 2\boldsymbol{j}) \times 20$
Scores B1M1A0.

A1

3

Note on Verification Method:
$\boldsymbol{v} = (-20\boldsymbol{i} - 8\boldsymbol{j}) + (3\boldsymbol{i} + 2\boldsymbol{j}) \times 20$ B1M1
$= (-20 + 60)\boldsymbol{i} + (-8 + 40)\boldsymbol{j}$
$= 40\boldsymbol{i} + 32\boldsymbol{j}$ A1
Similarly, verification to confirm acceleration from the two velocities is acceptable.

(ii) ($\mathbf{v}$ =) $(-20\mathbf{i} - 8\mathbf{j}) + (3\mathbf{i} + 2\mathbf{j})$

B1: Correct velocity vector.
*Note "****v*** *=" does not need to be seen.*

B1

1

(iii) ($\mathbf{v}$ =) $(3t - 20)\mathbf{i} + (2t - 8)\mathbf{j}$

M1: Velocity vector seen split into components. Condone omission of ***i*** *and* ***j***
Note: This can be implied by later working, such as the second line of this solution.

M1

$$(3t - 20)^2 + (2t - 8)^2 = 8^2$$

dM1: Equation based on speed of 8.

dM1

A1: Correct unsimplified equation.

A1

$$13t^2 - 152t + 400 = 0$$

A1: Simplified quadratic equation

A1

$$t = \frac{152 \pm \sqrt{152^2 - 4 \times 13 \times 400}}{2 \times 13}$$

dM1: Solving quadratic equation, to obtain two solutions.

dM1

$t = 4$ or $t = 7.69$

A1: Both correct solutions. Accept AWRT 7.7 or 7.6 or $\frac{100}{13}$.

Note: Using calculator to solve quadratic is acceptable.

A1

6

[14]

Q15.

(a) $\mathbf{v} = 9\mathbf{i} + 7\mathbf{j}$

B1: Correct initial velocity.

B1

1

(b) $\mathbf{v} = (9 - 0.01t)\mathbf{i} + (7 - 0.03t)\mathbf{j}$

M1: Writing $\mathbf{v}$ *in the form* $\mathbf{v} = \mathbf{u} + \mathbf{a}t$

M1

$= (9\mathbf{i} + 7\mathbf{j}) + (-0.01\mathbf{i} - 0.03\mathbf{j})t$

$\mathbf{a} = -0.01\mathbf{i} - 0.03\mathbf{j}$

A1: Correct acceleration.

A1

2

(c) $9 - 0.01t = -(7 - 0.03t)$

M1: Equation involving both ***i*** *and* ***j*** *components of their velocity.*

A1: Correct equation, from their acceleration.

M1A1

$t = \frac{16}{0.04} = 400$

A1: Correct time, from their acceleration.

A1

3

[6]

Q16.

(a) $10\mathbf{a} = 9\mathbf{i} + 12\mathbf{j}$

M1: Application of Newton's second Law with m = 10 in vector form.

M1

$\mathbf{a} = (0.9\mathbf{i} + 1.2\mathbf{j})$ m s^{-2}

A1: Correct acceleration.
If acceleration incorrect follow their value through for the rest of this question.

A1

2

(b) (i) $\mathbf{r}(5) = (2.2\mathbf{i} + 1\mathbf{j}) \times 5 + \frac{1}{2}(0.9\mathbf{i} + 1.2\mathbf{j}) \times 5^2$

M1: Use of constant acceleration to find position vector at t = 5, with $\boldsymbol{u} \neq 0\,\boldsymbol{i} + 0\,\boldsymbol{j}$.

M1

$= 22.25\mathbf{i} + 20\mathbf{j}$

*A1F: Correct position vector, for candidate's acceleration which must be a vector. Allow 22.3**i** + 20**j**.*

A1F

$d = \sqrt{22.25^2 + 20^2}$ = 29.9 metres

dM1: Calculation of distance from position vector. Must see + sign.

dM1

*A1F: Correct distance, for their acceleration. Accept 30 from 22.3**i** + 20**j**.*

A1F

4

(ii) $\mathbf{v} = (2.2\mathbf{i} + 1\mathbf{j}) + (0.9\mathbf{i} + 1.2\mathbf{j})t$

*M1: Use of constant acceleration equation to find an expression for **v**, with $\boldsymbol{u} \neq 0\,\boldsymbol{i} + 0\,\boldsymbol{j}$*

M1

*A1F: Correct **v** for their acceleration.*

A1F

2

(iii) $\mathbf{v} = (2.2 + 0.9t)\mathbf{i} + (1 + 1.2t)\mathbf{j}$

$2.2 + 0.9t = 1 + 1.2t$

$1.2 = 0.3t$

$t = 4$

*M1: Equation involving both **i** and **j** components of their velocity. Could have incorrect signs, for example*
$2.2 + 0.9t = -(1 + 1.2t)$.

M1

A1F: Correct equation.

A1F

A1F: Correct time, for their acceleration.

A1F

3

[11]

Q17.

(a) Using $\mathbf{F} = m\mathbf{a}$,

$400 \cos \frac{\pi}{2} t\,\mathbf{i} + 600t^2\,\mathbf{j} = 200\,\mathbf{a}$

M1

$\mathbf{a} = 2 \cos \frac{\pi}{2} t\,\mathbf{i} + 3t^2\,\mathbf{j}$

A1

2

(b) $\mathbf{v} = \int a\, \mathrm{d}t$

M1 for either $\int a\,\mathrm{d}t$ or 1 of 2 terms correct

M1

$= \frac{4}{\pi} \sin \frac{\pi}{2} t\,\mathbf{i} + t^3\,\mathbf{j} + \mathbf{c}$

*m1 for + **c***

A1m1

When $t = 4$, $\mathbf{r} = -3\mathbf{i} + 56\mathbf{j}$,
$64\mathbf{j} + \mathbf{c} = -3\mathbf{i} + 56\mathbf{j}$

m1

$\therefore\ \mathbf{c} = -3\,\mathbf{i} - 8\,\mathbf{j}$

$\therefore \mathbf{v} = (\frac{4}{\pi}\sin\frac{\pi}{2}t - 3)\mathbf{i} + (t^3 - 8)\mathbf{j}$

Do not accept $\frac{\frac{2}{\pi}}{2}$ *Accept 1.27 for* $\frac{4}{\pi}$

A1

5

(c) When particle is moving due west, northerly component is zero

M1

$\therefore t^3 - 8 = 0$

A1ft

$t = 2$

A1

3

(d) When $t = 2$, $\mathbf{v} = -3\mathbf{i} + 0\mathbf{j}$

B1ft

Speed of particle is 3 m s^{-1}

B1 for change –3 to +3

B1

2

[12]

Q18.

(a) Resultant = (6**i** – 3**j**) + (3**i** + 15**j**)
= 9**i** + 12**j**

M1: Summing the two vectors

M1

A1: Correct resultant

A1

2

(b) Magnitude $= \sqrt{9^2 + 12^2}$
= 15 N

M1: Finding magnitude with an addition sign.

M1

A1F: Correct magnitude based on their answer to part (a).

A1F

2

(c) $1.5m = 9$ or $2m = 12$

$m = 6$ kg $\quad m = 6$ kg

M1: Applying Newton's second law to one or both of the components.

M1

A1F: Correct mass, follow through their answer to part (a). Do not award this mark if vector division with 2 components has been used, eg

$\frac{9\mathbf{i} + 12\mathbf{j}}{1.5\mathbf{i} + 2\mathbf{j}} = 6$ *or 6**i** + 6**j** etc without a correct previous statement gives M0A0*

A1F

2

(d) (i) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j})t^2$

M1: Using a constant acceleration equation to find the position vector with $\mathbf{u} = 0\mathbf{i} + 0\mathbf{j}$

M1

A1: Correct position vector.

A1

2

(ii) $\mathbf{r} = \frac{1}{2}(1.5\mathbf{i} + 2\mathbf{j}) \times 2^2 = 3\mathbf{i} + 4\mathbf{j}$

$d = \sqrt{(3)^2 + (4)^2}$

M1: Finding the position vector when $t = 2$
($\mathbf{r} = (1.5\mathbf{i} + 2\mathbf{j}) \times 2 = 3\mathbf{i} + 4\mathbf{j}$ scores M0 unless it is clear how the 2 was obtained, possibly by a correct formula in (d) (i))

M1

$= \sqrt{25} = 5$

A1: Correct distance

A1
2
[10]

Q19.

(a) $\mathbf{v} = \dfrac{\mathrm{d}r}{\mathrm{d}t}$

M1

$\mathbf{v} = (e^{\frac{1}{2}t} - 8)\mathbf{i} + (2t - 6)\mathbf{j}$

***i** terms*

A1

***j** terms*

A1
3

(b) (i) When t = 3, **v** = –3.52**i**

Accept $(e^{\frac{3}{2}} - 8)\mathbf{i}$

B1

Speed is 3.52 m s^{-1}

3.5 does not give 2nd B mark

B1
2

(ii) West

B1
1

(c) $\mathbf{a} = \dfrac{1}{2}e^{\frac{1}{2}t}\mathbf{i} + 2\mathbf{j}$

M1A1

When t = 3, **a** = $\dfrac{1}{2}e^{\frac{3}{2}}$ **i** + 2**j** or 2.24**i** + 2**j**

A1
3

(d) Using **F** = m**a**:

*Accept **F** = 7**a***

M1

$\mathbf{F} = 7\left(\frac{1}{2}e^{\frac{3}{2}}\mathbf{i} + 2\mathbf{j}\right)$

$\therefore$ Magnitude of force is $7\left(\left(\frac{1}{2}e^{\frac{3}{2}}\right)^2 + 2^2\right)^{\frac{1}{2}}$

M1

F = 21.025
F = 21.0

Accept 21

A1

3

[12]

Q20.

(a) **v** = (–2**i** + 2**j**) + (0.25**i** + 0.3**j**) × 20

*M1: Finding velocity using **v** = **u** + **a**t*

M1

A1: Correct expression

A1

v = 3**i** + 8**j**

A1: Correct velocity in simplest form

A1

3

(b) – 2 + 0.25t = 0

*M1: One component equal to zero (either **i** or **j** component).*
A1: Correct equation

M1A1

t = 8 s

A1: Correct time

A1

$\mathbf{v} = (2 + 0.3 \times 8)\mathbf{j} = 4.4\mathbf{j}$

A1: Correct velocity

A1

4

(c) $\mathbf{r} = (-2\mathbf{i} + 2\mathbf{j}) \times 20 + \frac{1}{2}(0.25\mathbf{i} + 0.3\mathbf{j}) \times 20^2 + (9\mathbf{i} + 7\mathbf{j})$

M1: Finding position vector using a constant acceleration equation with or without the initial position with t = 20.
A1: Correct expression for position vector including initial position.

M1A1

OR

$\mathbf{r} = \frac{1}{2}((-2\mathbf{i} + 2\mathbf{j}) + (3\mathbf{i} + 8\mathbf{j})) \times 20 + (9\mathbf{i} + 7\mathbf{j})$

$\mathbf{r} = 19\mathbf{i} + 107\mathbf{j}$

A1: Correct position vector in simplest form.

A1

3

(d) $\mathbf{v}_{AVERAGE} = \frac{(19\mathbf{i} + 107\mathbf{j}) - (9\mathbf{i} + 7\mathbf{j})}{20}$

M1: Finding average velocity based on change of position. Subtraction of initial position must be seen or implied. Division by 8 scores M0

M1

$= \frac{10\mathbf{i} + 100\mathbf{j}}{20}$

$= 0.5\mathbf{i} + 5\mathbf{j}$

A1F: Correct average velocity. Follow through incorrect answers from part (c).

Allow $\frac{\mathbf{u} + \mathbf{v}}{2}$

A1F

2

[12]

Q21.

(a) $4\mathbf{i} = 5\mathbf{j} + 40\mathbf{a}$ **AG**

$\mathbf{a} = \dfrac{4\mathbf{i} - 5\mathbf{j}}{40} = 0.1\mathbf{i} - 0.125\mathbf{j}$

Forming a vector equation based on constant acceleration

M1

Correct equation

A1

Solving for ***a***

dM1

Correct ***a*** *from correct working*

A1

For $\dfrac{4\mathbf{i} - 5\mathbf{j}}{40}$ *on its own give M0*
Allow verification

4

(b) $\mathbf{r} = 5\mathbf{j} \times 40 + \dfrac{1}{2}(0.1\mathbf{i} - 0.125\mathbf{j}) \times 40^2$
$= 80\mathbf{i} + 100\mathbf{j}$

Finding position vector

M1

Correct expression

A1

Correct simplified result

A1

3

(c) (i) $\mathbf{v} = 5\mathbf{j} + (0.1\mathbf{i} - 0.125\mathbf{j})t$
$= 0.1t\mathbf{i} + (5 - 0.125t)\mathbf{j}$

Expression for ***v***

M1

Correct expression for ***v*** *seen or implied*

A1

$5 - 0.125t = -0.1t$
$5 = 0.025t$

Equating components, with or without a minus sign

dM1

Correct equation

A1

$t = \dfrac{5}{0.025} = 200$

Correct time.

A1

5

(ii) $\mathbf{v} = 0.1 \times 200\mathbf{i} + (5 - 0.125 \times 200)\,\mathbf{j}$
$= 20\mathbf{i} - 20\mathbf{j}$

Finding velocity using their time

M1

Correct velocity for their time

A1F

2

Note: Consistent use of u = 4i or 5i or a = 0.1i + 0.125j award method marks only.

[14]

Q22.

(a) $\mathbf{v} = \dfrac{\mathrm{d}r}{\mathrm{d}t}$

$\mathbf{v} = (3t^2 - 6t)\mathbf{i} + (4 + 2t)\mathbf{j}$

M1A1

2

(b) (i) $\mathbf{a} = (6t - 6)\mathbf{i} + 2\mathbf{j}$

M1A1ft

Using $\mathbf{F} = m\mathbf{a}$:
$\mathbf{F} = (18t - 18)\mathbf{i} + 6\mathbf{j}$

A1ft

3

(ii) When $t = 3$, $\mathbf{F} = 36\mathbf{i} + 6\mathbf{j}$

Magnitude is $\sqrt{36^2 + 6^2}$

M1

$= 36.5$

Accept $6\sqrt{37}$ *; ft from (b)(i)*

A1ft
2

(c) When **F** acts due north:
Component of **F** in the **i** direction is 0

M1

$18t - 18 = 0$
$t = 1$
ft from (b)(i)

A1ft
2

[9]

Q23.

(a) **F** = 5**j** + 8**i** – 7**j** = 8**i** – 2**j**
Adding the two forces. For incorrect answers, evidence of adding must be seen

M1

Correct resultant

A1
2

(b) $F = \sqrt{8^2 + 2^2} = \sqrt{68} = 8.25$ N
Finding magnitude (must see addition and not subtraction)

M1

Correct magnitude
Accept $2\sqrt{17}, \sqrt{68}$ *or AWRT 8.25 (eg 8.246)*

A1F
2

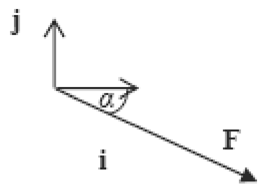
(c)


Diagram with force in the correct quadrant and with correct direction shown by an arrow.

B1

tan $\alpha = \frac{2}{8}$

Using trig to find angle: if tan, 8 in denominator; if sin or cos, 8.25 or their answer to part (b) in denominator

M1

$\alpha = 14.0°$

Correct angle
Accept 14.1 or 14 or AWRT 14.0 (eg 14.04)
M1 and A1 not dependent on B1

A1

3

[7]

Q24.

Use of column vectors is acceptable throughout this question.

(a) $\mathbf{v} = 20\mathbf{i} + (-0.4\mathbf{i} + 0.5\mathbf{j})t$

Use of constant acceleration equation to find expression for ***v***

M1

Any correct expression.

A1

2

(b) $\mathbf{v} = (20 - 0.4t)\mathbf{i} + 0.5t\mathbf{j}$

Simplifying ***v****. (May be implied.)*
(Missing brackets may be condoned if followed by correct working.)

M1

$20 - 0.4t = 0$

Putting ***i*** *component equal to zero*

m1

$t = \frac{20}{0.4} = 50$ seconds

Correct time
Candidates who are able to see the correct time without supporting working gain full marks.

Condone $\frac{20\mathbf{i}}{0.4\mathbf{i}} = 50$

A1

3

(c) $\mathbf{r} = 20\mathbf{i} \times t + \frac{1}{2}(-0.4\mathbf{i} + 0.5\mathbf{j}) \times t^2$

*Use of constant acceleration equation to find expression for **r***

M1

Any correct expression

A1

2

(d) (i) $\mathbf{r} = 20\mathbf{i} \times 100 + \frac{1}{2}(-0.4\mathbf{i} + 0.5\mathbf{j}) \times 100^2$

*Substituting t = 100 into their expression for **r** (dependent on M1 in part (c))*

m1

$= 2000\mathbf{i} - 2000\mathbf{i} + 2500\mathbf{j}$
$= 2500\mathbf{j}$

*Correct simplified position vector ie 2500**j***

A1

Therefore due north

*Conclusion that helicopter is due north provided their position vector is of the form k**j**, where k > 0*
Note if integration is used there is no need to prove that the constant is zero.
Note marks for (d) (i) can be awarded if part (c) scores zero.

A1

3

(ii) $\mathbf{v} = (20 - 0.4 \times 100)\mathbf{i} + 0.5 \times 100\mathbf{j}$

*Substituting t = 100 into their expression for **v** (dependent on M1 in part (a)) or use of other constant acceleration equation and their position vector (dependent on M1 in part (c))*

m1

$= -20\mathbf{i} + 50\mathbf{j}$

Correct simplified velocity

A1

$v = \sqrt{20^2 + 50^2} = 53.9$

Correct speed (Accept $10\sqrt{29}$ *)*

A1

Note marks for (d) (ii) can be awarded if part (a) scores zero.

3

[13]

Q25.

(a) $75\mathbf{i} = (5\mathbf{i} - 2\mathbf{j}) \times 10 + \frac{1}{2}\mathbf{a} \times 10^2$

Equation to find ***a*** *from* $\mathbf{r} = \mathbf{u}t + \frac{1}{2}\mathbf{a}t^2$

M1

Correct expression

A1

$\mathbf{a} = \dfrac{75\mathbf{i} - 50\mathbf{i} + 20\mathbf{j}}{50} = 0.5\mathbf{i} + 0.4\mathbf{j}$

AG *Correct* ***a*** *from correct working*

A1

3

(b) $\mathbf{r} = (5\mathbf{i} - 2\mathbf{j}) \times 8 + \frac{1}{2}(0.5\mathbf{i} + 0.4\mathbf{j}) \times 8^2$

Expression for ***r*** *using t = 8 with no extra terms*

M1

Correct expressions

A1

$= 56\mathbf{i} - 3.2\mathbf{j}$

Correct position vector

A1

3

(c) $\mathbf{v} = (5 + 0.5t)\mathbf{i} + (0.4t - 2)\mathbf{j}$

Expression for ***v****. Correct expression*

M1A1

$0.4t - 2 = 0$

$\mathbf{j}$ component equal to zero

dM1

$t = \frac{2}{0.4} = 5$

Correct t

A1

$\mathbf{r} = (5\mathbf{i} - 2\mathbf{i}) \times 5 + \frac{1}{2}(0.5\mathbf{i} + 0.4\mathbf{j}) \times 5^2$

Expression for $\mathbf{r}$ using t from $\mathbf{j}$ component equal to zero

dM1

$= 31.25\mathbf{i} - 5\mathbf{j}$
$= 31.3\mathbf{i} - 5\mathbf{j}$

Correct position vector

A1 6

[12]

Q26.

(a) $\mathbf{v} = 1.2\mathbf{i} + (-0.9 \sin t)\mathbf{j}$

Convincingly differentiation

M1A1 2

(b) (i) Speed = $\sqrt{1.2^2 + 0.9^2 \sin^2 t}$

M1A1 2

(ii) Maximum speed = $\sqrt{1.2^2 + 0.9^2}$

Use of sint = 1

m1

$= 1.5$

A1 2

[6]

Q27.

(a) $\mathbf{u} = 5\mathbf{i}$ or $\begin{bmatrix} 5 \\ 0 \end{bmatrix}$

Correct velocity

B1

1

(b) $\mathbf{v} = 5\mathbf{i} + (-0.2\mathbf{i} + 0.25\mathbf{j})t$

*Use of constant acceleration equation, with **u** and **a** not zero*

M1

Correct velocity
*M1A0 for using 5**j** or just 5*

A1

OR

$\mathbf{v} = \begin{bmatrix} 5 - 0.2t \\ 0.25t \end{bmatrix}$

2

(c) $5 - 0.2t = 0$

Easterly component zero

M1

Correct equation

A1

$t = \dfrac{5}{0.2} =$ 25 seconds

Correct t

A1

3

(d) $\mathbf{r} = 5\mathbf{i} \times 25 + \frac{1}{2}(-0.2\mathbf{i} + 0.25\mathbf{j}) \times 25^2$

Use of constant acceleration equation with t from part (c)

M1

Correct expression based on t from part (c)

A1F

$= 62.5\mathbf{i} + 78.125\mathbf{j}$

Correct simplification CAO

A1

$\theta = \tan^{-1}\left(\frac{62.5}{78.125}\right)$

Using tan to find the angle

dM1

Correct expression based on t from part (c), with correct two values (either way)

A1F

$= 038.7°$

Correct angle
Accept 38.6° or 039°

A1

OR

$\mathbf{r} = \frac{1}{2}(5\mathbf{i} + 6.25\mathbf{j}) \times 25$

(M1)(A1F)(A1)

$\theta = \tan^{-1}\left(\frac{5}{6.25}\right) = 038.7°$

(dM1)(A1F)(A1)

6

[12]

Q28.

(a) $\mathbf{v} = 4\mathbf{i} + (-3\mathbf{i} + 12\mathbf{j})t$

M1

use of $\boldsymbol{v} = \boldsymbol{u} + \boldsymbol{a}t$

A1

2

(b) $t = 0.5$, $\mathbf{v} = 2.5\mathbf{i} + 6\mathbf{j}$

Ft 2 terms and t subs

B1ft

Speed = $\sqrt{(2.5^2 + 6^2}$

2 terms

M1

Speed = 6.5 ms[1]

ft 2 terms

A1ft

3

[5]

Q29.

(a) $\mathbf{F} = 12\cos t\mathbf{i} - 30\sin t\mathbf{j}$

Use of **F** = *m***a**

M1

$\mathbf{F}(0) = 6 \times 2\mathbf{i}$ so $F = 12$ N

Correct **F**

A1

Correct magnitude

A1

3

(b) $\mathbf{v} = \int 2\cos t\,dt\,\mathbf{i} + \int -5\sin t\,dt\,\mathbf{j}$

Integrating

M1

$= (2\sin t + c_1)\mathbf{i} + (5\cos t + c_2)\mathbf{j}$

Correct **i** *component*

A1

Correct **j** *component*

A1

Both of above with or without constants

$\mathbf{v}(0) = 2\mathbf{i} + 10\mathbf{j} \Rightarrow c_1 = 2,\ c_2 = 5$

Finding constants of integration

dM1

$\mathbf{v} = (2\sin t + 2)\mathbf{i} + (5\cos t + 5)\mathbf{j}$

Correct final answer

A1

5

[8]

Q30.

(a) $\mathbf{d} = 3\mathbf{i} - 6\mathbf{j}$

*Accept ± **d** or displacement of 3, 6 shown on a diagram*

B1

$3\mathbf{i} - 6\mathbf{j} = (\mathbf{i} - 2\mathbf{j})t$

Or equivalent method for t

Accept ratio of vectors leading directly to ± 3

M1

$t = 3$

CAO

A1

3

(b) (i) $\mathbf{r} = (\mathbf{i} - 2\mathbf{j}) \times 4 + \frac{1}{2} \times 2\mathbf{j} \times 16$

M1

full method for vector expression giving change in position

*for correct subs (give 4**i** + 8**j**)*

A1

$+6\mathbf{i} - 4\mathbf{j}$

M1

$= 10\mathbf{i} + 4\mathbf{j}$

ft slip provided obtain vector expression

*(**u** = 0 gives 6**i** + 12**j**)*

A1F

4

(ii) $A(3,2)$ $C(10,4)$

$\mathbf{d} = 7\mathbf{i} + 2\mathbf{j}$

*Attempt to find vector **AC** or **CA***

(using candidate's C)

M1

$|\mathbf{d}| = \sqrt{7^2 + 2^2}$

$AC = \sqrt{53} = 7.28$

*ft **d** provided two non-zero components*

Accept $\sqrt{53}$

A1F

2

[9]

Q31.

(a) $\mathbf{v} = (6t^2 - 2t)\mathbf{i} + (1 - 12t^2)\mathbf{j}$

differentiating both components

M1

one component correct

A1

second component correct

A1

3

(b) (i) $\mathbf{v}\left(\frac{1}{3}\right) = \left(\frac{6}{9} - \frac{2}{3}\right)\mathbf{i} + \left(1 - \frac{12}{9}\right)\mathbf{j} = -\frac{1}{3}\mathbf{j}$

*substituting the value for t into their **v***

M1

correct velocity

A1

2

(ii) Travelling due south

correct description (Follow through from $\mathbf{v} = \pm k\mathbf{j}$ *)*

A1ft

1

(c) **a** = (12t – 2)**i** – 24t**j**

differentiating their velocity

M1

a(4) = 46**i** – 96**i**

correct acceleration at time t

A1

correct acceleration at t = 4

A1

3

(d) $\mathbf{F} = 6(46\mathbf{i} - 96\mathbf{j}) = 276\mathbf{i} - 576\mathbf{j}$

apply Newton's second law correctly

M1

$F = \sqrt{276^2 + 576^2} = 639 \text{ N}$

finding magnitude

M1

correct magnitude

A1

3

or

$a = \sqrt{46^2 + 96^2} = 106.45$

$F = 6 \times 106.45 = 639 \text{ N}$

[12]

Q32.

(a) $-\mathbf{i} + \mathbf{j} = 2\mathbf{i} - \mathbf{j} + 10\mathbf{a}$

Use of velocity equation

M1

$\mathbf{a} = -0.3\mathbf{i} + 0.2\mathbf{j}$

Correct equation

A1

*Correct **a***

A1

3

(b) $\mathbf{r} = (2\mathbf{i} - \mathbf{j})t + \frac{1}{2}(-0.3\mathbf{i} + 0.2\mathbf{j})t^2 + 20\mathbf{i}$

Use of constant acceleration equation for position

M1

$= (2t - 0.15t^2 + 20)\mathbf{i} + (-t + 0.1t^2)\mathbf{j}$

*Correct **i** component*

A1

*Correct **j** component*

A1ft

3

ft incorrect acceleration

(c) (i) $\mathbf{r}(20) = (2 \times 20 - 0.15 \times 20^2 + 20)\mathbf{i} + (-20 + 0.1 \times 20^2)\mathbf{j}$
$= 0\mathbf{i} + 20\mathbf{j}$

*Substituting t = 20 into their expression for **r***

M1

so due north of origin

Correct conclusion from correct working

A1

2

(c) (ii) $\mathbf{v}(20) = 2\mathbf{i} - \mathbf{j} + 20(-0.3\mathbf{i} + 0.2\mathbf{j}) = -4\mathbf{i} + 3\mathbf{j}$

$\mathbf{v}(20) = \sqrt{4^2 + 3^2} = 5 \text{ ms}^{-1}$

Finding velocity at t = 20

M1

Correct velocity

A1

Finding magnitude

m1

Correct speed

A1ft

4

ft incorrect acceleration

[12]

Q33.

(a) $\mathbf{v} = 4\mathbf{j} + (3\mathbf{i} - 5\mathbf{j})\, t$

*M1: Use of **v** = **u** + **a**t and*
***u** ≠ 0 or Integration*
A1: Correct expression

M1A1

2

(b) $\mathbf{v} = 3t\mathbf{i} + (4 - 5t)\,\mathbf{j}$

*M1: **j** component of velocity equal to zero*

M1

$4 - 5t = 0$
$t = 0.8$ s
A1: Correct t

A1
2

(c) $\mathbf{v} = 12\mathbf{i} - 16\mathbf{j}$
M1: Finding velocity when t = 4

M1

$\mathbf{v} = \sqrt{12^2 + 16^2} = 20$ **AG**
A1: Correct velocity

A1

dM1: Finding the magnitude

dM1

A1: Correct speed from correct working

A1
4
[8]

Q34.

(a) Ball is a particle/no spin
B1: One assumption

B1

No air resistance/Only gravity or weight
B1: Second assumption

B1
2

(b) (i) $24.5\,t - 4.9t^2 = 0$
M1: Equation for vertical motion with height zero

M1

$\left(t = 0 \text{ or } t = \frac{24.5}{4.9} = 5s\right)$ **AG**
A1: Correct equation

A1

dM1: Solving for t

dM1

A1: Correct time from correct working

A1

4

(b) (ii) $R = 10 \times 5 = 50$ m

M1: Use of horizontal component of velocity to find the range

M1

A1: Correct range

A1

2

(c) $20 = 10\,t$

M1:Horizontal equation

M1

$t = 2$

A1: Time to reach wall

A1

$h = 24.5 \times 2 - 4.9 \times 2^2 = 29.4$ m

dM1:Vertical equation for height with u = 24.5 and a negative acceleration

dM1

A1: Correct height

A1

4

(d) No change as acceleration and initial velocity do not change with the mass

B1: No change

B1

B1: Explanation

B1

2

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Q35.

Answers in (a), (b) and (c)(i) must be vectors

(a) $\mathbf{a} = 4t\mathbf{i} + 3\mathbf{j}$

Must be vector, both i and j present

B1 1

(b) $\mathbf{v} = 2t^2\mathbf{i} + 3t\mathbf{j}$

Integration of acceleration attempted

M1

*ft **a**, both **i** and **j** present*

A1F 2

(c) (i) $\mathbf{r} = \int_0^t \left(2t^2\mathbf{i} + 3t\mathbf{j}\right)\mathrm{d}t$

Integration attempted

M1

$= \frac{2t^3}{3}\mathbf{i} + \frac{3t^2}{2}\mathrm{j}$

*A1 each term. Condone **i** and **j** missing.*
–1 for constants present,
–1 for both terms unsimplified

A1A1 3

(ii) $\left.\begin{array}{l}\mathbf{v}\text{ parallel to }\mathbf{i}+\mathbf{j}\\ 2t^2 = 3t\end{array}\right\}$

M1

$t = \frac{3}{2}$

*Solve, ft **v** with vector components*

A1F

$\mathbf{r} = \frac{9}{4}\mathbf{i} + \frac{27}{8}\mathbf{j}$

A1F

Distance $= \sqrt{\left(\left(\frac{9}{4}\right)^2 + \left(\frac{27}{8}\right)^2\right)}$

*Magnitude of **r** in any form*

M1

= 4.06 (4.056)

*Numerical answer, ft candidate's **r** and t*

A1F

5

[11]